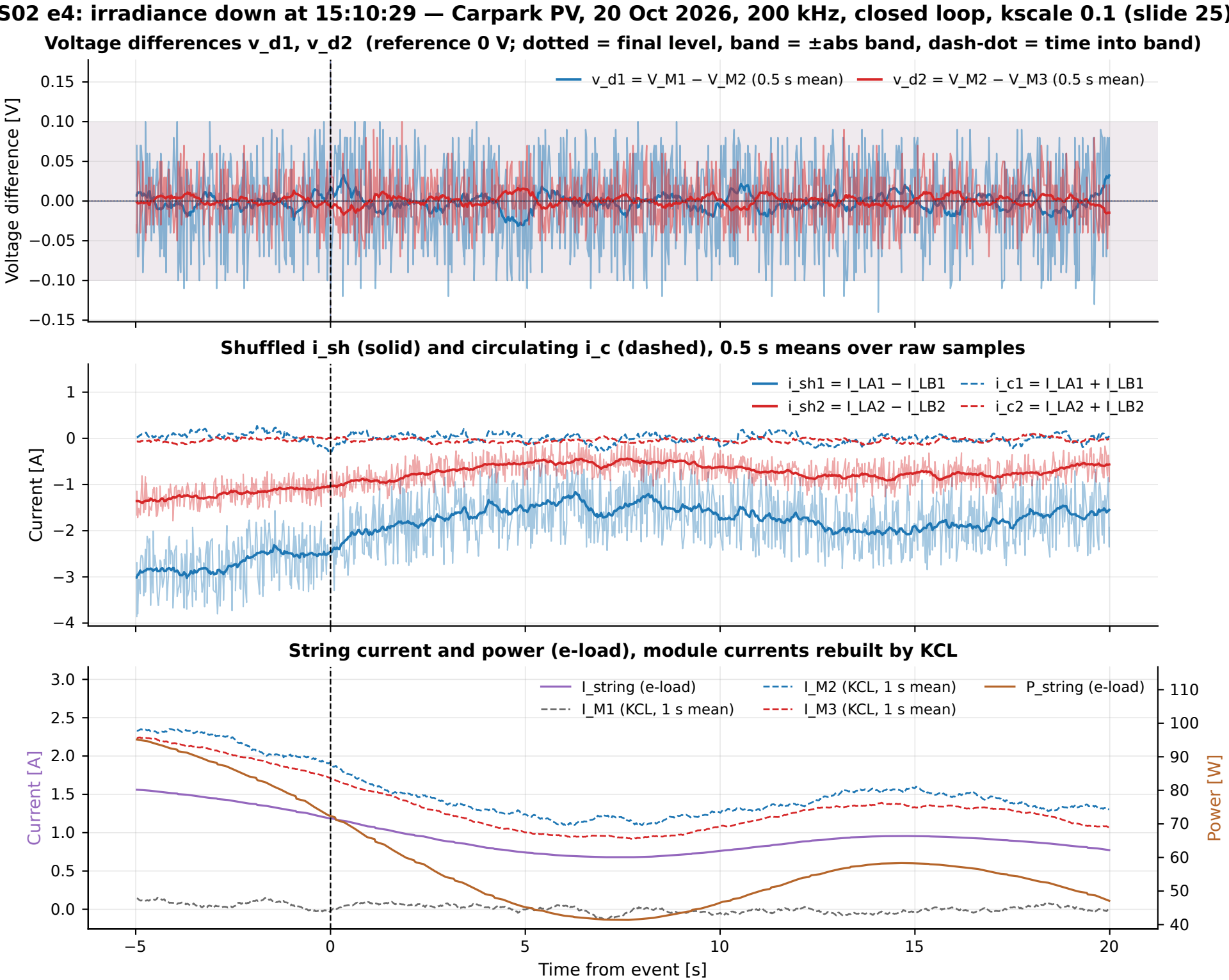

S02 e4: irradiance down at 15:10:29 — Carpark PV, 20 Oct 2026, 200 kHz, closed loop, kscale 0.1 (slide 25)
Voltage differences v_d1, v_d2 (reference 0 V; dotted = final level, band = ±abs band, dash-dot = time into band)
v_d1 = V_M1 − V_M2 (0.5 s mean)
v_d2 = V_M2 − V_M3 (0.5 s mean)
Voltage difference [V]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Shuffled i_sh (solid) and circulating i_c (dashed), 0.5 s means over raw samples
i_sh1 = I_LA1 − I_LB1
i_c1 = I_LA1 + I_LB1
i_sh2 = I_LA2 − I_LB2
i_c2 = I_LA2 + I_LB2
Current [A]
1
0
−1
−2
−3
−4
String current and power (e-load), module currents rebuilt by KCL
I_string (e-load)
I_M2 (KCL, 1 s mean)
P_string (e-load)
I_M1 (KCL, 1 s mean)
I_M3 (KCL, 1 s mean)
Current [A]
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Power [W]
110
100
90
80
70
60
50
40
−5
0
5
10
15
20
Time from event [s]
P_string -24.1 W over 4 s | telemetry 44 Hz: times shorter than one sample (23 ms) are not resolved